

A Balanced Covering Pair of Size $\sqrt{n}$ and Polynomial Magnitude: Resolving a Difference-Set Question for the Smoothness Covering Decomposition

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Abstract

A companion note to the Smoothness Covering Decomposition showed how to cover $[1, n]$ by divisibility with a set D of size $\pi(n)(1+o(1))$ and elements of magnitude $\exp(O(\sqrt{n}))$, and asked (Remark 3 there) for the least K such that a *balanced* pair (S, T) with $|S| = |T| = K$ covers $[1, n]$ by nonzero differences — every $i \leq n$ dividing some $s - t$ with $s \in S, t \in T, s \neq t$. That remark cautioned the balanced realization looked infeasible for $K \ll |D|$. We show the opposite: an elementary two-scale grid gives a balanced pair with $K = \lceil \sqrt{n+1} \rceil = (1+o(1))\sqrt{n}$ and all elements of magnitude at most $n + O(\sqrt{n})$. This improves the trivial realization simultaneously on *both* axes — size $\sqrt{n}$ versus $\pi(n)$, and magnitude $O(n)$ versus $\exp(O(\sqrt{n}))$. We also observe that minimizing K alone is degenerate ($K = 1$ via a single least-common-multiple element, at magnitude $\exp(O(n))$), so the meaningful object is the size–magnitude trade-off; the grid occupies one clean corner of it, and the shape of the full curve (in particular, any nontrivial lower bound on K at polynomial magnitude) is left open. The finite claims were verified by direct enumeration. This note has not been externally refereed.

1 Setup

We work with the divisibility-covering notion of the companion decomposition.

Definition 1. A pair of finite sets $S, T \subset \mathbb{Z}$ is a *covering pair* for $[1, n]$ if for every integer i with $1 \leq i \leq n$ there exist $s \in S, t \in T$ with $s \neq t$ and $i \mid (s - t)$. It is *balanced* if $|S| = |T|$; we write $K = |S| = |T|$. Its *magnitude* is $\max\{|x| : x \in S \cup T\}$.

The nonzero requirement $s \neq t$ is essential: without it the difference 0 is divisible by every i , and $S = T = \{0\}$ would “cover” $[1, n]$ vacuously. With it, the problem is the genuine one raised in Remark 3 of the companion note.

The trivial realization $S = D, T = \{0\}$ from a covering set D gives $K = |D|$; for the decomposition’s D that is $\pi(n)(1+o(1))$ at magnitude $\exp(O(\sqrt{n}))$. Remark 3 there worried that a balanced realization with $K \ll |D|$ is obstructed, because prescribing the $|D|$ specific values of D as differences imposes $|D|$ constraints on the $2K$ values of $S \cup T$. The resolution below is to abandon reproducing D ’s specific values: a balanced pair need not realize D at all, only cover $[1, n]$ afresh.

2 The construction

Theorem 1 (Balanced covering pair). *For every $n \geq 1$ let $m = \lceil \sqrt{n+1} \rceil$ and set*

$$S = \{0, 1, 2, \dots, m-1\}, \quad T = \{0, -m, -2m, \dots, -(m-1)m\}.$$

Then (S, T) is a balanced covering pair for $[1, n]$ with $K = m = (1 + o(1))\sqrt{n}$ and magnitude $(m - 1)m < n + O(\sqrt{n})$.

Proof. For $s = a \in S$ and $t = -bm \in T$ ($0 \leq a, b \leq m-1$) the difference is $s - t = a + bm$. As a ranges over $\{0, \dots, m-1\}$ and b over $\{0, \dots, m-1\}$, the value $a + bm$ takes *every* integer in $\{0, 1, \dots, m^2 - 1\}$ exactly once (base- m representation). Since $m = \lceil \sqrt{n+1} \rceil$ gives $m^2 \geq n + 1$, the difference set $\{s - t\}$ contains $\{1, 2, \dots, n\}$. Hence for each $i \leq n$ the value i itself is a nonzero difference, and trivially $i \mid i$; the pair covers $[1, n]$. Both sets have m elements, so $K = m = (1 + o(1))\sqrt{n}$. The largest magnitude is $|(m-1)m| = (m-1)m$, and with $m < \sqrt{n+1} + 1$ this is below $n + O(\sqrt{n})$. $\square$

Remark 1 (Two-axis improvement). The pair of Theorem 1 improves the trivial realization on both axes at once: size $\sqrt{n}$ against $\pi(n) \sim n/\ln n$, and magnitude $O(n)$ against the decomposition's $\exp(O(\sqrt{n}))$. It is therefore not merely a balanced substitute for D but a strictly cheaper covering object in both parameters. The obstruction of Remark 3 in the companion note does not bind because the construction never reproduces the specific elements of D ; it covers $[1, n]$ directly through the m^2 differences of an $m \times m$ grid.

3 Minimizing K alone is degenerate; the trade-off is the real object

Proposition 1. For every n the pair $S = \{0\}$, $T = \{\text{lcm}(1, \dots, n)\}$ is a balanced covering pair for $[1, n]$ with $K = 1$. Its magnitude is $\text{lcm}(1, \dots, n) = \exp((1 + o(1))n)$.

Proof. The single nonzero difference is $-\text{lcm}(1, \dots, n)$, which every $i \leq n$ divides. The magnitude estimate is the prime number theorem for the second Chebyshev function, $\log \text{lcm}(1, \dots, n) = \psi(n) \sim n$. $\square$

Thus the bare question “least K ” has answer 1 and is not the interesting one: the cost has simply migrated into magnitude, from $\exp(O(\sqrt{n}))$ up to $\exp(O(n))$. The meaningful object is the *size–magnitude trade-off*: the least K achievable subject to a magnitude bound M . Three points on this curve are now known:

- (i) $K = 1$, $M = \exp(O(n))$ (Proposition 1);
- (ii) $K = \pi(n)$, $M = \exp(O(\sqrt{n}))$ (decomposition's D);
- (iii) $K = \sqrt{n}$, $M = O(n)$ (Theorem 1).

Point (iii) dominates (ii) in both coordinates, so (ii) is not on the lower boundary of the trade-off. The genuinely open direction is the boundary near (iii):

Question 1. At polynomial magnitude $M = n^{O(1)}$, is $K = \Theta(\sqrt{n})$ optimal, or can a balanced covering pair have $K = o(\sqrt{n})$? Equivalently, is there a lower bound $K = \Omega(\sqrt{n})$ for covering pairs of polynomially bounded magnitude?

A lower bound here cannot come from a crude counting of differences alone: covering needs only a *multiple* of each $i \leq n$ among the $\leq K^2$ differences, not i itself, and the relaxation to multiples is what makes small- n optima dip below $\lceil \sqrt{n} \rceil$ (see Section 4). Whether that relaxation buys an asymptotic improvement at polynomial magnitude, or is washed out, is what the question asks.

4 Verification

The construction of Theorem 1 was checked by direct enumeration: for $n \in \{16, 100, 1000, 10^4\}$ the pair (S, T) leaves no $i \leq n$ uncovered, and the measured $K/\sqrt{n}$ is 1.25, 1.10, 1.01, 1.01 respectively, converging to 1. The magnitudes $(m-1)m$ are 20, 110, 992, 10100 against $n = 16, 100, 1000, 10^4$, confirming the $O(n)$ bound. Proposition 1 was checked for $n \leq 500$: the single element $\text{lcm}(1, \dots, n)$ is divisible by every $i \leq n$.

Exhaustive minimum- K search (nonzero differences, values bounded by $2n$) shows the multiple-relaxation genuinely dips below $\lceil \sqrt{n} \rceil$ at small n : the least K is 2 up to $n = 10$ and 3 up to $n = 20$, whereas $\lceil \sqrt{n} \rceil$ reaches 3 already at $n = 5$ and 4 at $n = 10$. For instance $S = \{0, 2\}, T = \{16, 20\}$ covers $[1, 10]$ with $K = 2$ (differences $\{-14, -16, -18, -20\}$, whose divisors include every $i \leq 10$). These are the data motivating the open question; they do not settle the asymptotic rate, since the search is exponential and was carried out only for $n \leq 24$.

Acknowledgement of scope

The upper bound of Theorem 1 is complete and elementary. The lower-bound side of the trade-off — and hence the exact optimal K at polynomial magnitude — is not resolved here.

References

- [1] E. Schultz, *A Smoothness Covering Decomposition: Covering $[1, n]$ by Divisibility with One Dedicated Element per Large Prime Power*, 2026 (companion note; Remark 3 poses the balanced-pair question resolved here).